K. J. Somaiya School of Engineering, Mumbai · Somaiya Vidyavihar University
Artificial Intelligence (216U41C501) · Lab CA Mini-Project · TY CCE (EXTC) · Sem V · 2025-26

ClauseGuard: Detecting Unfair Clauses in Terms & Conditions with NLP

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Abstract

People routinely accept terms and conditions without reading them. ClauseGuard is an NLP system that reads a document and flags clauses that are potentially unfair to the user. We trained and compared three classifiers on the UNFAIR-ToS dataset (9,414 sentences from 50 online terms of service, 8 unfair-clause types): TF-IDF with Logistic Regression, TF-IDF with a Linear SVM, and a fine-tuned DistilBERT transformer. On the held-out test set the best model (DistilBERT (fine-tuned)) reached a macro F1 of 0.802 and an F1 of 0.797 for separating fair from unfair clauses. The model is deployed in a web app that accepts PDFs, scanned documents, Word files, photos, camera shots, links and pasted text, highlights risky clauses with plain-English explanations and computes a 0–100 risk score.

1. Problem statement

Terms of service, rental agreements and offer letters are long and written in legal language, so most people agree without reading them. A Deloitte survey of 2,000 US consumers found that 91% accept legal terms without reading them (97% of 18–34 year olds) [4]. In a study by Obar and Oeldorf-Hirsch, 98% of students missed a clause that gave away their first-born child [5], and reading every privacy policy a person meets in a year would take about 76 working days [6]. Users therefore unknowingly accept arbitration instead of courts, one-sided termination and changes, liability waivers, auto-renewals and penalties. The goal is a tool that automatically points out such clauses and quantifies the overall risk before the user agrees.

2. Dataset

We used UNFAIR-ToS from the LexGLUE benchmark [2], built on the CLAUDETTE corpus [1]. It contains 9,414 sentences from 50 terms of service (e.g. of large online platforms), labelled by legal experts with zero or more of 8 unfair types: limitation of liability, unilateral termination, unilateral change, content removal, contract by using, choice of law, jurisdiction and arbitration. We kept the official splits: 5,532 training, 2,275 validation and 1,607 test sentences. Only 11.4% of training sentences are unfair, so the data is strongly imbalanced; 63 sentences carry two labels. We framed the task as 9-class classification (Fair + 8 types); multi-label training sentences were duplicated once per label.

3. Methodology

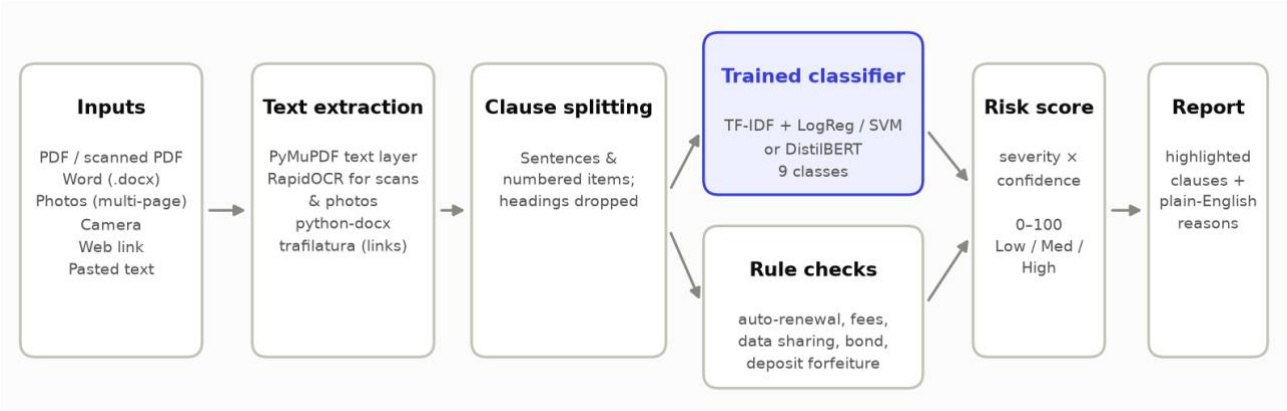


Figure 1. ClauseGuard pipeline.

- **Text extraction:** PyMuPDF for PDFs; pages with no text layer are rendered and read with RapidOCR (also used for photos and the camera, with up-scaling and a word-segmentation fix for merged words); python-docx for Word files; trafilatura for web pages.

- **Clause splitting:** a rule-based splitter on sentence ends and numbered items that drops headings.
- **Classical models:** TF-IDF features (word 1–2-grams + character 3–5-grams, sub-linear TF) with Logistic Regression and with a Linear SVM (probabilities via Platt-style calibration). Both use class_weight = balanced to counter the imbalance.
- **Deep learning:** DistilBERT (distilbert-base-uncased) fine-tuned for 3 epochs on CPU (AdamW, lr 5e-05, batch 16, max length 128, linear warm-up) with a cross-entropy loss weighted by the square root of the balanced class weights (a deliberately milder form of balancing, so rare classes count more without dominating); the epoch with the best validation macro F1 was kept.
- **Rule checks:** 5 transparent keyword rules for patterns outside the dataset that students often meet (auto-renewal, hidden fees, data sharing, bond / lock-in, deposit forfeiture), always labelled separately from model predictions.
- **Risk score:** each finding contributes risk = severity of its type × model confidence; score = 100 × (1 − e^(−Σ risk / 2.5)), giving Low (0–29), Medium (30–60) or High (61–100).

4. Results

All metrics are on the official test split, which was not used for training or model selection.

Model	Accuracy	Macro P	Macro R	Macro F1	Unfair-only macro F1	Fair vs unfair F1	Train time
TF-IDF + Logistic Regression	0.946	0.736	0.845	0.775	0.750	0.802	6 s
TF-IDF + Linear SVM	0.955	0.874	0.714	0.779	0.754	0.800	2 s
DistilBERT (fine-tuned) (best)	**0.948**	**0.769**	**0.847**	**0.802**	**0.780**	**0.797**	**69 min**

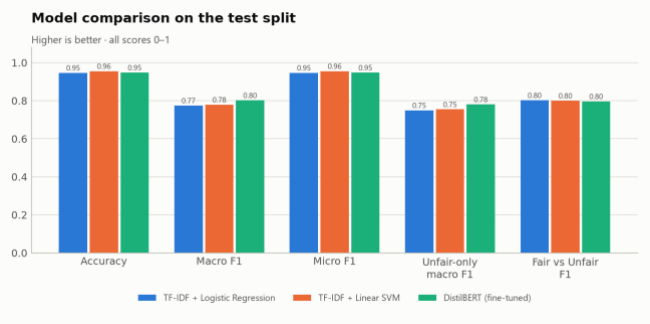


Figure 2. Model comparison (test split).

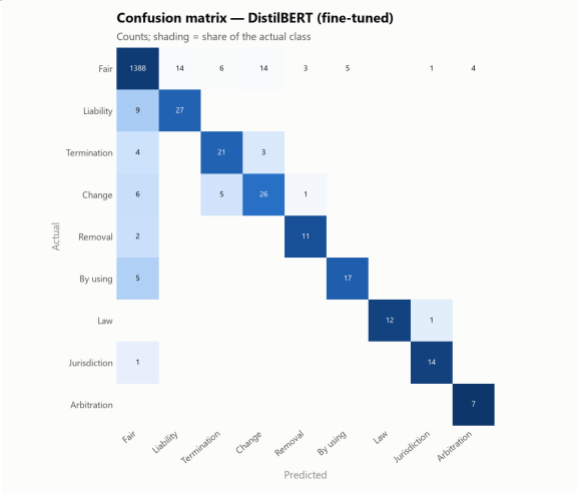


Figure 3. Confusion matrix, DistilBERT (fine-tuned).

Accuracy is high for every model because about 89% of test sentences are fair, so macro F1 (all 9 classes weighted equally) is the main metric. DistilBERT (fine-tuned) was selected for the app because it has the best validation macro F1 (0.744). Per class, it is strongest on Choice of law (F1 0.96) and weakest on Unilateral change (F1 0.64, 38 test sentences). For DistilBERT (fine-tuned), 47 fair sentences were flagged as unfair (false alarms) and 27 unfair sentences were missed. The most common mix-up between unfair types was Unilateral change predicted as Unilateral termination (5 sentences).

5. Application

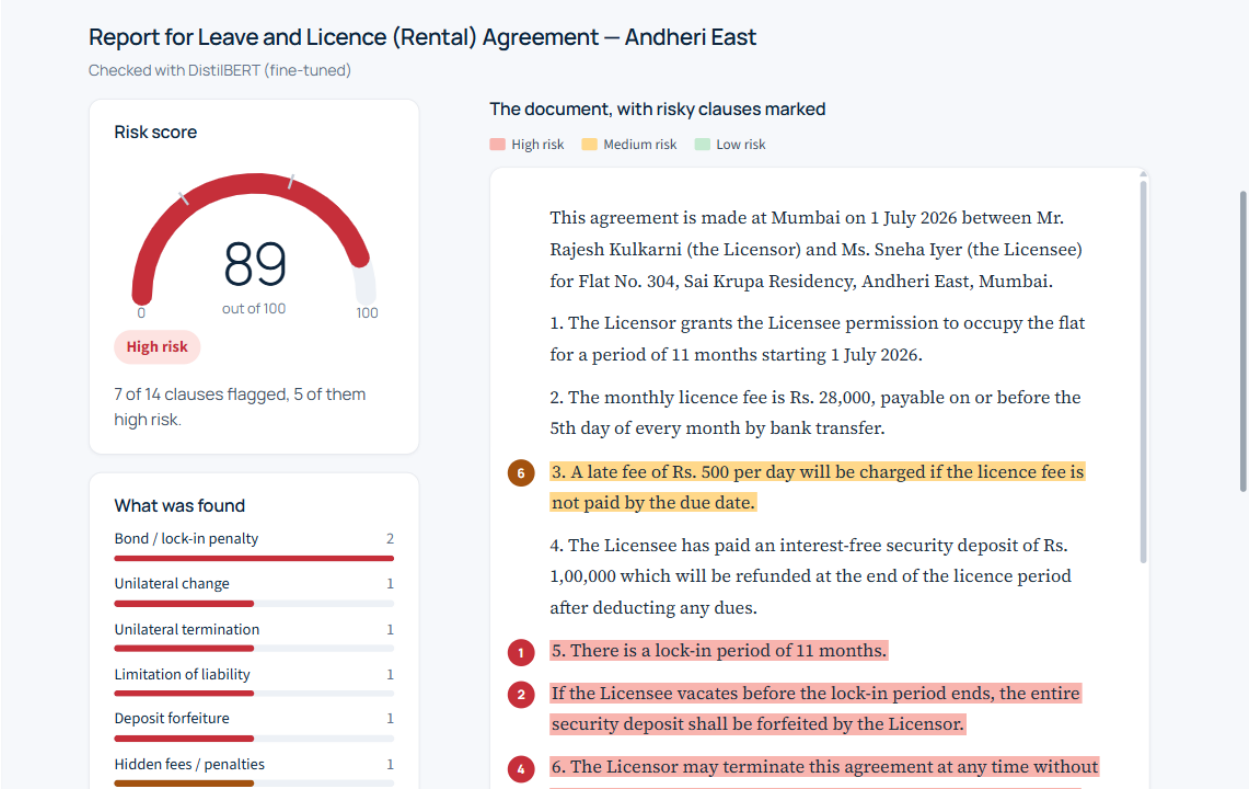


Figure 4. ClauseGuard web app: risk score, category summary and the document with risky clauses highlighted.

The Streamlit app offers tabs for PDF, Word, multiple images, camera, link and pasted text, three built-in sample documents (app terms, a rental agreement and an internship offer letter), a downloadable HTML report and a model-performance page. It runs on a laptop and on phones over the same Wi-Fi (HTTPS mode for the camera). PDF, scanned PDF, Word, multi-image, link and pasted-text inputs were tested end-to-end with automated browser tests; the camera path shares the image OCR code and is covered by a phone-style photo test. A pytest suite covers extraction, clause splitting, rules, scoring, HTML escaping and the app's behaviour.

6. Conclusion and future scope

ClauseGuard shows that a model trained on a modest labelled dataset can reliably flag unfair clauses and make long legal documents understandable in seconds. Its main limitation is domain: it learned from English online terms of service, so it is less accurate on Indian rental agreements and offer letters, where the rule checks help. Future work: collect and label Indian contracts (rent, employment, loan apps), support Hindi and Marathi, fine-tune a legal-domain model such as LegalBERT, suggest fairer rewordings of risky clauses, and ship a browser extension that scans terms before the user clicks "I agree". ClauseGuard is not legal advice.

References

[1] M. Lippi et al., "CLAUDETTE: an automated detector of potentially unfair clauses in online terms of service," Artificial Intelligence and Law, vol. 27, pp. 117–139, 2019.

[2] I. Chalkidis et al., "LexGLUE: A Benchmark Dataset for Legal Language Understanding in English," in Proc. ACL, 2022.

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[5] J. A. Obar and A. Oeldorf-Hirsch, "The biggest lie on the Internet: ignoring the privacy policies and terms of service policies of social networking services," Information, Communication & Society, 23(1), 2020.

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Source code

The complete project (code, trained models, dataset scripts, tests and samples) is on GitHub: https://github.com/Abutallah24/clauseguard